

# How to read the software

fe-large-displacement

Abaqus is several programs that share a name. This repository is a thin layer on top of them. Confusing the parts is how a three-day job is spent on a keyword CAE never wrote.

## 1 Abaqus/CAE versus the solvers

Abaqus/CAE builds a model. It does not solve it. Two solvers do:

- Abaqus/Standard — implicit. Equilibrium iterations each increment.
- Abaqus/Explicit — explicit dynamics. A stable time increment, no iteration.

CAE writes an input file (`.inp`). The solver reads that file. CAE accepts arguments it does not implement and omits keywords whose values match the default. The deck is the ground truth. Every model here writes the `.inp`, checks it, and only then submits.

## 2 Standard versus Explicit

|                        | Standard              | Explicit                          |
|------------------------|-----------------------|-----------------------------------|
| Procedure              | <b>*Static</b> (here) | <b>*Dynamic, Explicit</b>         |
| Failure mode           | no convergence        | element distortion                |
| User material          | UMAT                  | VUMAT                             |
| Pore-pressure elements | C3D8P                 | none on ordinary solids           |
| ALE                    | limited               | production tool                   |
| CEL                    | none                  | Explicit only                     |
| Precision              | double                | set <code>DOUBLE_PLUS_PACK</code> |

Standard is the right tool until the soil starts to fail in a large-displacement sense. Explicit is the right tool after that, provided the run is shown to be quasi-static (`ALLKE/ALLIE`).

UMAT and VUMAT are different calling conventions for the same constitutive idea. A closed-form Mohr–Coulomb return can be wrapped. A model with substepping cannot be wrapped blindly. Check a user material on one element before a penetration job.

## 3 ALE and CEL are solver features

They are not separate products.

ALE keeps the mesh topology and moves nodes relative to the material. Explicit ALE is what this repository uses in production. Standard ALE exists on geometrically nonlinear **\*Static** steps. It has one smoothing algorithm, no initial mesh sweeps, and Lagrangian domains only. The implicit-ALE model shows that path. It is not the tool for skirt penetration.

CEL lets material flow through a fixed Eulerian mesh. The structure stays Lagrangian. CEL is Explicit only. There is no implicit CEL.

## 4 What this repository is

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|                |                                               |
|----------------|-----------------------------------------------|
| examples/      | one complete script per formulation           |
| constitutive/  | UMAT/VUMAT sources, each with its own licence |
| documentation/ | this PDF, and the formulations companion      |

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## 5 How a job is launched

```
set FLD_WORKDIR=C:\abq\run
set FLD_SUBMIT=0
abaqus cae noGUI=examples/implicit.py
```

FLD\_SUBMIT=0 writes the deck without solving. Start there.

When a user subroutine is compiled, launch through `abaqus.bat`, never `ABQLauncher.exe` directly. Paths with spaces break the Fortran compile.

## 6 The habit that catches silent failures

Write the deck. Read it. Then submit. Three failures produce a normal ODB and no warning: an ALE domain that never sweeps, a CEL domain with no void for heave, and a VUMAT returning NaN (Explicit integrates it to the end).

On a new machine, write the decks with FLD\_SUBMIT=0 before submitting. The code that changes from implicit to CEL is `documentation/formulations.pdf`.

## 7 Clausen Mohr–Coulomb

Use of `constitutive/mohr-coulomb-clausen/` requires citation of all three papers. BibTeX is in that directory as `references.bib`.

1. Clausen, J., Damkilde, L. and Andersen, L. (2006). Efficient return algorithms for associated plasticity with multiple yield planes. *International Journal for Numerical Methods in Engineering* 66(6), 1036–1059. <https://doi.org/10.1002/nme.1595>
2. Clausen, J., Damkilde, L. and Andersen, L. (2007). An efficient return algorithm for non-associated plasticity with linear yield criteria in principal stress space. *Computers & Structures* 85(23), 1795–1807. <https://doi.org/10.1016/j.compstruc.2007.04.002>
3. Clausen, J., Damkilde, L. and Andersen, L. V. (2015). Robust and efficient handling of yield surface discontinuities in elasto-plastic finite element calculations. *Engineering Computations* 32(6), 1722–1752. <https://doi.org/10.1108/EC-01-2014-0008>